

ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

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Final Exam Question on **GA**:

Q2b. Solution:

Generation 1 - Mutation Process:

i	Population	Site	Mutated Population	Value x_i	Fitness $f(x_i)$
1	001.0000	2	011.0000	3.0000	3.9841
2	010.0000	3	011.0000	3.0000	3.9841
3	011.0100	4	011.1100	3.7500	15.7480
4	100.0000	5	100.0100	4.2500	1.7746
5	101.1000	6	101.1010	5.6250	0.2214
6	110.1000	7	110.1001	6.5625	0.1066

Fitness Sum = **39.7922**; Fitness Average = **6.6320**

```

format short

% Mutation Process

cross_values = [1.00 2.00 3.45 3.80 5.50 6.50];

k = 1:length(cross_values);

pop = [];
for i = 1:length(cross_values)
    d2b = string(fix(rem(cross_values(i)*pow2(-(3-1):4),2)));
    dec = join(d2b(1:3), "");
    frac = join(d2b(4:7), "");
    pop(end+1) = join([dec, frac], ".");
end

pop;

sites = [2 3 4 5 6 7];

mut_pop = [011.0000 011.0000 011.1111 011.1000 101.1010 110.1001];

mut_values = [3.00 3.00 3.9375 3.5 5.625 6.5625];

mut_fitness = [];
for i = 1:length(mut_values)
    mut_fitness(end+1) = 1 / (mut_values(i)^2 - 7*mut_values(i) + 12.25 + 0.001);
end

% [k' pop' sites' mut_pop' mut_values' mut_fitness']
T = array2table([k' pop' sites' mut_pop' mut_values' mut_fitness'], ...
    "VariableNames", {'i' 'Pop' 'Sites' 'Mut Pop' 'Values (xi)' 'Fitness f(xi)'})

mut_fitness_sum = sum(mut_fitness)
mut_fitness_avg = sum(mut_fitness) / length(mut_fitness)

```

Decimal to Binary

```
function binaryFraction = decimalToBinaryFraction(decimalFraction)
    integerPart = floor(decimalFraction);
    fractionalPart = decimalFraction - integerPart;
    binaryInteger = dec2bin(integerPart);
    binaryFraction = '';
    for i = 1:16
        fractionalPart = fractionalPart * 2;
        if fractionalPart >= 1
            binaryFraction = [binaryFraction '1'];
            fractionalPart = fractionalPart - 1;
        else
            binaryFraction = [binaryFraction '0'];
        end
        if fractionalPart == 0
            break;
        end
    end
    binaryFraction = [binaryInteger '.' binaryFraction];
end

% Example usage:
x = 3.75;
y = decimalToBinaryFraction(x);
disp(['Decimal of ', num2str(x), ' is in binary ', y]);
```

Binary to Decimal

```
function decimalFraction = binaryToDecimalFraction(binaryFraction)
    parts = strsplit(binaryFraction, '.');
    decimalInteger = bin2dec(parts{1});
    decimalFractional = 0;
    for i = 1:length(parts{2})
        if parts{2}(i) == '1'
            decimalFractional = decimalFractional + 2^(-i);
        end
    end
    decimalFraction = decimalInteger + decimalFractional;
end

% Example usage:
y = '11.11';
x = binaryToDecimalFraction(y);
disp(['Binary ', y, ' is in decimal ', num2str(x)]);
```